

Contexts, Modalities, and Quantum Logic: A Critical Comparison of CSM with the Birkhoff–von Neumann Tradition

Karl Svozil¹

¹*Vienna, Austria*

(Dated: June 10, 2026)

The Contexts–Systems–Modalities (CSM) interpretation treats quantum properties as definite modalities belonging to a system in a macroscopic context. This note compares that view with the quantum-logical method introduced by Birkhoff and von Neumann and later sharpened by Gleason, Specker, and Kochen. Both approaches replace one global Boolean event algebra by locally Boolean but mutually incompatible contexts. The comparison is not seamless, however. Standard quantum logic is a type-I projection-lattice formalism, whereas recent CSM emphasizes macroscopic, irreversible contexts and type II/III operator algebras. Moreover, the firefly logic L_{12} shows that contextual Boolean pasting alone is compatible with classical two-valued states. Thus CSM’s primitives do not by themselves force specifically quantum contextuality; the specifically quantum content enters through the Hilbert-space or operator-algebraic representation, together with Gleason–Uhlhorn and Kochen–Specker-type constraints.

I. INTRODUCTION

Grangier’s recent note presents CSM as a non-FAPP, ontological interpretation of quantum mechanics [1]. Its primitives are contexts, systems, and modalities. A modality is a definite, repeatable outcome, but it is not a property of the isolated system alone; it belongs to a system together with a definite macroscopic context. The paper further stresses that real contexts are not merely finite ancillary quantum systems: detectors and records are macroscopic, dissipative, and irreversible, motivating an operator-algebraic description of contexts.

This program has a natural point of contact with the quantum logic of Birkhoff and von Neumann [2]. There, experimental yes–no propositions are represented by closed subspaces, or equivalently by projections, of a Hilbert space. The resulting structure is an orthocomplemented, orthomodular lattice and, in general, not a Boolean algebra because distributivity fails. Boolean logic survives only locally, inside a maximal compatible context. At this level the comparison is immediate: CSM gives an ontological reading of a contextual structure that quantum logic had already isolated algebraically.

The comparison nevertheless has to be made with some care. CSM is not simply “quantum logic plus ontology.” The Birkhoff–von Neumann construction is internal to a projection lattice such as $\text{Proj}(\mathcal{H})$, normally the lattice of a type-I Hilbert-space representation. Recent CSM instead wants the macroscopic context to be described by sectors, irreversible records, and type II or type III von Neumann algebras [23, 24]. Those objects are not just larger Boolean blocks pasted inside the same type-I lattice. Thus the relation between CSM and quantum logic is both continuous and tense: the vocabulary of contexts and modalities is naturally quantum-logical, but the macroscopic algebraic setting changes the meaning of some of the older lattice-theoretic notions.

II. QUANTUM LOGIC AND CSM

For a Hilbert space $\mathcal{H}$, the quantum-logical event structure is

$$\mathcal{L}(\mathcal{H}) = \text{Proj}(\mathcal{H}), \quad (1)$$

partially ordered by inclusion of subspaces, with orthocomplement $P^\perp$. Its nonclassical feature is not merely that distributivity can fail,

$$P \wedge (Q \vee R) \neq (P \wedge Q) \vee (P \wedge R), \quad (2)$$

but that this failure occurs in the controlled setting of an orthomodular lattice. Orthomodularity is the residual order-theoretic structure left after classical distributivity is abandoned; it is closely tied to projection, compatibility, and the Sasaki-type conditional operations used in the quantum-logical account of measurement [6, 7]. More concretely, when an ideal test of Q gives the affirmative result, the Lüders rule

$$\rho \mapsto \frac{Q\rho Q}{\text{Tr}(\rho Q)} \quad (3)$$

is the state-space counterpart of this projection structure; in lattice language the corresponding Sasaki projection sends P to $Q \wedge (P \vee Q^\perp)$. Thus CSM’s operational talk of changing context meets quantum logic precisely at the point where projection, repeatability, and conditionalization are tied together. Within a maximal commuting family of projections, however, the structure is Boolean. Quantum logic is therefore not a rejection of ordinary logic inside a given experiment; it is the rejection of one all-embracing Boolean algebra for all possible experiments. Specker famously illustrated this with the parable of an overprotective seer who provides answers to individual questions but forbids simultaneous answers to incompatible ones [4].

This last point is often formulated more flexibly by using partial Boolean algebras or pasted Boolean blocks.

In that language, one keeps Boolean operations only inside compatible contexts and does not insist that all joins and meets be globally meaningful. Historically, this also brings the comparison closer to Specker- and Kochen–Specker-style partial algebras than to a single global orthomodular lattice, because CSM operationally needs only context-wise Boolean structure plus cross-context identifications, not a fully defined global meet and join for arbitrary propositions. This is especially close to the CSM picture, because a context in CSM also fixes a set of mutually exclusive modalities, while changes of context need not be represented as moves inside one classical event space.

Gleason’s theorem supplies the probabilistic side of the type-I projection structure: in Hilbert spaces of dimension at least three, every frame function that is additive on each orthonormal resolution of the identity has the Born form

$$\mu(P) = \text{Tr}(\rho P). \quad (4)$$

It is common to call this assumption probabilistic non-contextuality, since the value assigned to a projector does not depend on the basis in which it is embedded. This use should be distinguished from Kochen–Specker noncontextuality, which concerns 0–1 truth-value assignments rather than probability measures. Specker’s theorem and the Kochen–Specker theorem provide the corresponding truth-value obstruction: suitable partial Boolean or projection structures admit no global assignment of 0’s and 1’s respecting the exclusivity constraints inside all contexts [4, 5]. Specker’s 1960 result is best read as a partial-Boolean obstruction, whereas the Kochen–Specker theorem gives the familiar Hilbert-space no-go theorem for noncontextual hidden variables. The familiar two-dimensional exception should be kept in mind: neither the original Gleason theorem nor Kochen–Specker obstruction applies directly to qubits, so any fully general CSM reconstruction must either treat the qubit case separately or appeal to enlarged measurement structures such as POVMs or composite embeddings [14, 15].

CSM translates this algebraic situation into ontological language. In the ordinary type-I Hilbert-space description, a context corresponds to a maximal set of mutually orthogonal projectors resolving the identity. A modality then corresponds naturally to an atom, or definite proposition, in that Boolean block. A change of context is a change to another, generally incompatible, Boolean block. The additional CSM claim is that the context is not only a basis in a calculation but a real macroscopic arrangement. Hence the definiteness of outcomes is not derived from unitary dynamics on one finite Hilbert space; it is built into the ontology of system–context modalities and, in Grangier’s recent formulation, into the algebraic description of irreversible macroscopic sectors [1, 23].

At exactly this point the comparison becomes delicate. In type II and type III factors there are no minimal projections; type III factors additionally admit no nonzero normal semifinite trace. In particular, the normal-state

structure is very different from the atomic type-I case and does not support pure normal states corresponding to lattice atoms. Thus the lattice-theoretic phrase “modality as an atom of a Boolean block” cannot be carried unchanged into the macroscopic algebraic regime. If CSM identifies modalities in that regime with sectors, folia, or equivalence classes of representations, then “modality” has shifted from an atomic proposition to an algebraic superselection or representation-theoretic notion. A sector, however, is not a minimal yes–no event but a representation-theoretic class of states; identifying it with a single definite outcome therefore requires an additional interpretive step. That shift may be physically well motivated, but it is not part of the original Birkhoff–von Neumann lattice picture and should not be hidden by vocabulary.

III. SIMILARITIES, DIFFERENCES, AND TENSIONS

The most important similarity is the replacement of global Booleanity by local Booleanity. Both approaches allow ordinary alternatives inside a single context. What they deny is the classical extrapolation from one Boolean context to a universal Boolean algebra containing all possible alternatives. For quantum logic this denial is expressed by non-distributivity, orthomodularity, and the partial pasting of Boolean blocks. For CSM it is expressed by the thesis that modalities are meaningful only relative to a specified system–context pair.

A second similarity is the role of probability. In the Birkhoff–von Neumann tradition, probabilities are states on the lattice of propositions. Gleason’s theorem then shows that, under frame-function additivity, such states are represented by density operators. In CSM, the same formal fact is read more physically: probabilities occur when a modality prepared in one context is interrogated in another incompatible context. Thus probabilities are not ignorance about a hidden Boolean state space, but transition weights between contexts. In the rank-one case these are literally the usual Gleason–Born values: for modalities represented by $P_\psi = |\psi\rangle\langle\psi|$ and $P_\phi = |\phi\rangle\langle\phi|$,

$$\text{Tr}(P_\psi P_\phi) = |\langle\phi|\psi\rangle|^2, \quad (5)$$

which is exactly the transition weight CSM posits between the two modalities. Grangier’s program further claims that Born’s rule can be recovered from contextual quantization, using ingredients such as extracontextuality, Uhlhorn-type symmetry considerations, and Gleason additivity [8–11]. This is also the point of Grangier’s later discussion of Kolmogorovian censorship and predictive incompleteness [12]: within a fixed context, or commuting subalgebra, the probabilities are ordinary Kolmogorov probabilities, while the genuinely quantum question is how these context-wise classical assignments are glued into one probability law on the projection lattice. In that language, the quantum state is predictively

TABLE I. Compressed comparison of quantum logic and CSM.

Aspect	Quantum logic	CSM	Critical point
Primitive	Propositions $P \in \text{Proj}(\mathcal{H})$	Modalities of system plus context	Same formal role in type I; different ontology
Context	Boolean block or commuting set	Macroscopic arrangement	CSM makes the context physically real
Classical domain	Boolean logic inside one block	Definite modalities inside one context	Essentially the same local Booleanity
Nonclassical feature	No global Boolean algebra	No context-free list of properties	Largely the same obstruction, rephrased ontologically
Probabilities	Gleason frame functions on projections	Transition weights between contexts	Formally linked, not competing accounts
Truth values	Kochen–Specker obstruction	No global intrinsic modality assignment	Same no-go fact in lattice and ontic vocabularies
Measurement update	Orthomodular/projection structure	Context change and repeatability	Natural meeting point, but rarely explicit in CSM
Macroscopic side	Historically type-I focused; extendable to general von Neumann algebras	Irreversible sectors, type II/III algebras	Atom/modality translation becomes problematic
Bell/EPR setting	Composite projection structure, but no apparatus ontology	Must treat separated macroscopic contexts	Relation between contextuality and nonlocality remains to be specified

incomplete until the measurement context is specified. This reinforces the quantum-logical reading of CSM: local classicality is unproblematic, and the nonclassical content lies in the cross-context projection structure. The same separation appears in finite non-Boolean event structures: local normalization inside each context is weaker than single-valued cross-context gluing, and even a glued admissible weight is still not thereby classical, Born-rule realizable, or physically realized [22].

This is also where a real limitation appears. The derivation of Born’s rule does not follow from the bare words context, system, and modality. It requires a representation in which modalities can be identified across contexts, transition probabilities have the required additivity or symmetry properties, and the relevant Hilbert-space or operator-algebraic machinery is already in place. It is worth being explicit about the logical order. Uhlhorn’s theorem characterizes the symmetry maps of an *already given* orthogonality structure on rays, and Gleason’s theorem presupposes the projection lattice of a Hilbert space of dimension at least three. Thus the CSM derivation should be read as: *given* that modalities are represented by rays with the quantum orthogonality relation, the only admissible transition weights are the Born ones. The substantive quantum commitment is the ray representation itself; once it is in place, Born’s rule is close to forced. This is precisely the asymmetry the firefly example exhibits at the qualitative level. In that sense the claimed “derivation” of Born’s rule is conditional rather than autonomous: once a Hilbert-space representation, orthogonality-preserving context transformations, and Gleason-type additivity are granted, the Born form follows; but those assumptions already encode

much of the specifically quantum structure. CSM’s distinctive ideas of extracontextuality and extravalence are introduced precisely to bridge this gap: they let the same modality be recognized across different contexts while retaining contextual objectivity [9, 11, 13]. From the quantum-logical side, this is not merely an interpretive gloss; it is an additional structural assumption.

The decisive difference is therefore not that quantum logic has mathematics while CSM has ontology. Rather, quantum logic is deliberately austere about measurement apparatuses: it classifies propositions, compatibilities, probability measures, and valuation obstructions, but it does not by itself say what a macroscopic record is or why such a record is unique. CSM makes precisely this extra interpretive move. It turns a context from a maximal commuting family into a real macroscopic arrangement and treats a modality as an actually possessed property of the system in that arrangement.

This difference explains the operator-algebraic emphasis in Grangier’s recent formulation. The original quantum-logical scheme can be developed inside the projection lattice of an ordinary Hilbert space, although quantum-logical methods can also be generalized to projection lattices of von Neumann algebras. CSM claims that the type-I Hilbert-space picture is not enough for the macroscopic side of measurement: real contexts are stable, irreversible, and thermodynamic. Hence type II or type III algebraic structures enter not as mathematical decoration, but as a proposed description of the physical definiteness of the apparatus. Grangier’s own illustration is instructive here. He couples each Stern–Gerlach output channel to an infinite reservoir of two-level systems prepared in a KMS state, so that, in the

intended thermodynamic-limit idealization, the two post-measurement macrostates are represented by inequivalent, ideally disjoint, GNS representations of the quasi-local algebra [1]. This is a clean realization of the claim that “one result only” is encoded by inequivalence rather than by collapse. But it also makes the lattice translation problem concrete: the relevant “modality” is now a folium of states (a disjoint representation), not an atom of a projection lattice. For any finite reservoir, however, the apparatus algebra remains type I and the relevant states lie in one Hilbert-space representation; disjointness and strict irreversibility appear only in the infinite-volume limit.

Finite-dimensional decoherence estimates make this contrast sharper rather than removing it. In a large environmental Hilbert space, typical branch records can be quasi-orthogonal, with overlaps of order $e^{-S/2}$ when the accessible dimension is $d_{\text{eff}} \sim e^S$ [21]. This gives a powerful FAPP explanation of the invisibility of macroscopic interference. But without a specified system–environment split and interaction Hamiltonian it does not uniquely select a pointer basis; and even when einselection supplies stable pointer states, it does not turn an improper mixture into a proper one. It therefore supports the distinction drawn here: ordinary Hilbert-space decoherence supplies record stability, while CSM’s ontology of macroscopic contexts is an additional interpretive commitment.

A natural question is how this contextual reading bears on Bell-type nonlocality. In the quantum-logical and partial-Boolean tradition, the Kochen–Specker obstruction is state-independent and concerns a single system, whereas Bell inequalities are state-dependent and concern spacelike-separated wings. CSM treats a Bell experiment as a single bipartite context: the joint measurement arrangement at both wings fixes one set of mutually exclusive modalities [12]. In that reading the “no global valuation” fact of Kochen–Specker and the “no local hidden-variable” fact of Bell are closely related failures of one classical global probability space, though Bell’s theorem adds the specifically spacetime assumptions of local causality and measurement independence. What the firefly example shows for contextuality has a Bell analogue: bare contextual vocabulary does not by itself produce a Tsirelson-bounded violation; the quantitative violation again comes from the Hilbert-space transition structure.

IV. THE FIREFLY LOGIC L_{12}

The firefly logic L_{12} is a useful calibration point because it separates weak contextuality from Kochen–Specker contextuality. It consists of two three-atom Boolean contexts,

$$C_1 = \{a_1, a_2, a_3\}, \quad C_2 = \{a_3, a_4, a_5\}, \quad (6)$$

pasted along the common atom a_3 . Thus the Hasse diagram has twelve elements and five atoms; operationally it is often described by a firefly in a box observed from two sides, with the shared atom corresponding to the “no flash” alternative [16–19].

Quantum-logically, L_{12} is already non-Boolean: the alternatives to the shared atom depend on which test is performed. In CSM language, this is a toy model of two macroscopic contexts with one modality identifiable across both contexts. It therefore captures the grammar of contextual modalities in a minimal way.

But L_{12} is not a Kochen–Specker logic. Because the two contexts share only one atom, it admits many two-valued states, indeed enough to separate the atoms. Explicitly, either $v(a_3) = 1$ and all other atoms receive value 0, or $v(a_3) = 0$ and one chooses exactly one of $\{a_1, a_2\}$ and exactly one of $\{a_4, a_5\}$ to receive value 1. Thus L_{12} has five dispersion-free states, separating its five atoms. One can locate L_{12} on a finer scale. Full Kochen–Specker sets forbid even a single global $\{0, 1\}$ valuation (state-independent contextuality), while weaker constructions only forbid noncontextual reproduction of specific quantum statistics (state-dependent contextuality). L_{12} fails both: it admits a full separating set of two-valued states. It is therefore not merely “not state-independent KS”; it is classically embeddable outright. It is nondistributive and contextual as a pasted empirical logic, yet still classically representable. This is stronger than the modest claim that contexts alone are insufficient for Kochen–Specker contextuality. It shows that the CSM primitives, taken by themselves, are compatible with a classical hidden-variable representation. The specifically quantum content must be imported through the Hilbert-space or operator-algebraic structure that CSM adds to those primitives.

The firefly example therefore marks a limit on the explanatory autonomy of CSM. A structure may have contexts, modalities, repeatability, and non-distributive pasting while still admitting global two-valued states and no Gleason-type Born rule. Partition-logic analyses sharpen the same point: the logico-empirical incidence structure can support classical probabilities from the convex hull of dispersion-free states as well as Born-type probabilities from faithful orthogonal graph representations, so the structure alone does not select the probability theory [20]. To obtain the stronger CSM conclusion one needs more: finite-dimensional Hilbert space with dimension at least three, or an appropriate operator-algebraic analogue; extracontextual identification of modalities across contexts; and the Gleason–Uhlhorn constraints on transition probabilities. The example is useful precisely because it keeps these ingredients apart.

V. CONCLUSION

CSM can be read as a descendant of the Birkhoff–von Neumann insight, but not as a smooth ontological completion of it. Quantum logic already supplies much of the formal grammar: local Boolean contexts, non-Boolean global structure, orthomodularity, Gleason-type probability measures, and Specker–Kochen restrictions on global truth assignments. CSM accepts much of this grammar but changes its interpretation: properties are modalities of system–context pairs, and the context is a physically real macroscopic arrangement.

The advantage of quantum logic is economy and precision. It separates the projection structure, probability theory, and valuation obstructions without adding a detailed ontology of measurement. The advantage of CSM is precisely that it attempts to supply such an ontology, including the irreversibility and unicity of macroscopic contexts. But the attempt brings three obligations. It

must explain how type-I atomic modalities relate to type II/III sectors; it must make clear which extra assumptions, beyond the bare CSM primitives, recover Born’s rule and exclude classical two-valued models; and it must say how contextual objectivity reproduces Bell violations while specifying the status of spacelike separated contexts and avoiding a hidden global context.

The firefly logic L_{12} marks the lower boundary of this comparison. It displays contextual Boolean pasting, but not the full quantum obstruction. Thus quantum logic gives the algebraic grammar of quantum propositions; CSM proposes how that grammar is instantiated in actual measurements. The proposal is promising, but its force comes from the additional Hilbert-space and operator-algebraic structure, not from contextual vocabulary alone. The real question is therefore not whether CSM is contextual—so is quantum logic—but which additional structures turn contextuality into specifically quantum ontology.

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